

SMART CONTRACT RISK

Over the past decade, crypto-focused products, primarily exchanges, have repeatedly been hacked.¹ Whereas many of these situations happened because of poor security practices, they demonstrate an important point: software is uniquely vulnerable to hacks and developer malpractice. Blockchains can remove traditional financial risks, such as counterparty risk, with their unique properties, but DeFi is built on code. This software foundation gives attackers a larger surface than the threat vectors of traditional financial institutions. As discussed previously, public blockchains are open systems. After the code is deployed, anyone can view and interact with it on a blockchain. Because it is often responsible for storing and transferring blockchain native financial assets, it introduces a new, unique risk. This new attack vector is termed *smart contract risk*.

DeFi's foundation is public computer code known as a smart contract. First introduced by Nick Szabo in his 1997 paper,² its implementation is new to mainstream engineering practice and thus solutions for smart contract bugs and programming errors are still under development. The recent hacks of DForce and bZx demonstrate the fragility of smart contract programming, and auditing firms like Quantstamp, Trail of Bits, and Peckshield are emerging to fill this gap in best practices and smart contract expertise.³

Smart contract risk can take the form of a logic error in the code or an economic exploit in which an attacker can